

ATHUL RAVI

Assistant Professor of Chemistry · Founding Head, Centre for Biomaterials and Sustainable Chemistry (CBSC)

Chinmaya Vishwavidyapeeth (Deemed University), Piravom, Ernakulam, Kerala, India

Doctoral Research Scholar, Amrita Vishwa Vidyapeetham, Coimbatore

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PROFESSIONAL SUMMARY

Assistant Professor of Chemistry and founding head of the Centre for Biomaterials and Sustainable Chemistry (CBSC) — the first science research centre established at Chinmaya Vishwavidyapeeth — with an active doctoral research programme in biomaterials for tissue regeneration and wound healing. Research spans 3D-printed biopolymer hydrogel scaffolds, cerium-oxide nanozymes, phytochemical extraction and phytopharmaceutical validation, integrated with computational and AI-driven methods for materials design and biological network analysis. Committed to evidence-grounded, mechanistically rigorous science, with a distinctive research identity built at the intersection of Indian Knowledge Systems (Ayurveda and phytochemistry) and modern materials science. Experienced in institutional research leadership, curriculum development aligned with NEP 2020, and securing industry, government and AICTE-funded programmes.

RESEARCH INTERESTS

- Biomaterials and scaffolds for wound healing and tissue regeneration, including 3D-printed biopolymer hydrogels
- Cerium-oxide nanozymes and redox-active nanomaterials for modulation of the wound micro-environment
- Phytochemical extraction, phytopharmaceuticals, and scientific validation of traditional (Ayurvedic) medicine
- Sustainable and biomass-derived materials; valorisation of agricultural by-products
- AI- and machine-learning-guided materials design, cheminformatics, and biological network analysis
- Drug discovery, molecular docking, and 3D-QSAR; novel and targeted drug-delivery systems

EDUCATION

Ph.D. in Chemistry (Pursuing) · Amrita Vishwa Vidyapeetham, Coimbatore, India *2020 – Present*
Multifunctional Materials and Analysis Laboratory (AMMALab), Department of Sciences, School of Physical Sciences. Major area: biomaterials for tissue regeneration; minor area: natural-products isolation.

M.Sc. in Chemistry · Amrita Vishwa Vidyapeetham, Amritapuri Campus, Kerala, India *2014 – 2016*
CGPA 7.11/10. Thesis: in-silico discovery of Aurora Kinase-B inhibitors by 3D-QSAR and molecular docking.

B.Sc. in Chemistry · S.B. College, Changanacherry (M.G. University), Kerala, India *2011 – 2014*

PROFESSIONAL EXPERIENCE

Assistant Professor of Chemistry · Chinmaya Vishwavidyapeeth (Deemed University), Ernakulam, Kerala *2025 – Present*

- Teach chemistry and lead curriculum development aligned with NEP 2020, AICTE, OBE and NBA frameworks, including interdisciplinary courses integrating materials science, green chemistry, biomedical technology and Indian Knowledge Systems (IKS).

Founding Head (formerly Director), CBSC · Chinmaya Vishwavidyapeeth, Ernakulam, Kerala *Mar 2025 – Present*

- Conceived and established the Centre for Biomaterials and Sustainable Chemistry — the first science research centre at Chinmaya Vishwavidyapeeth — funded through CSR support from Suven Pharmaceutical Ltd.
- Served as founding Director (Mar 2025 – Apr 2026); on re-designation of the post, continued to lead the Centre as Head from Apr 2026.
- Direct the Centre's research agenda on biomaterials for diabetic wound healing, tissue regeneration and drug delivery, integrating IKS as a substantive research thread alongside modern materials science and AI methods.
- Supervise student research projects spanning biomaterials, machine-learning-guided composite design and AI-based biological network analysis; lead grant development and industry engagement.

Assistant Director, Centre for IKS–CKS Integration and Innovation · Chinmaya Vishwavidyapeeth, Ernakulam, Kerala *Feb 2025 – Dec 2025*

- Held institutional administrative responsibility for integrating Indian Knowledge Systems and Contemporary Knowledge Systems into research and academic initiatives across the university.

Doctoral Research Scholar / Research Assistant · AMMALab, Amrita Vishwa Vidyapeetham, Coimbatore *Jan 2023 – Present*

- Design and characterise biomaterials — hydrogel scaffolds, 3D scaffolds, silica-based composites and nanocomposites — for tissue-regeneration applications.
- Conduct phytochemical extraction and in-vitro biological evaluation; analyse data and present findings to the research group and scientific audiences.

Teaching Assistant · Amrita Vishwa Vidyapeetham, Coimbatore *Aug 2020 – Dec 2022*

- Delivered laboratory instruction and demonstrations for undergraduate and postgraduate students; prepared course materials and assessed student work.

Assistant Professor · MGM College of Engineering and Technology, Ernakulam, Kerala *Aug 2018 – Jul 2020*

- Taught chemistry to engineering students; served as examination in-charge, chief hostel warden, vice-chair of anti-ragging and discipline committees, and member of the students' grievance cell and IQAC; managed laboratory procurement.

Assistant Professor · Ilahia School of Science and Technology, Ernakulam, Kerala *Jan 2018 – Jul 2018*

- Instructed chemistry courses and supported laboratory sessions, curriculum development and programme-outcome evaluation.

Chemist Trainee · BPCL Kochi Refinery, Kerala *Jul 2016 – Apr 2017*

- Analysed petroleum products against production and quality specifications; supported analytical troubleshooting and laboratory operations.

PUBLICATIONS

1. P. P. Darshitha, **A. Ravi**, P. Lasya, M. Menon, G. Sivasubramanian, et al. "Foxtail millet husk as an innovative biomass in the preparation of silica–silver composite with antimicrobial and free-radical scavenging activities." *Materials Today: Proceedings*, vol. 66, pp. 1830–1836, 2022.
2. G. Sivasubramanian, N. J. Anagha, P. Devika, P. Archana, **A. Ravi**, et al. "Dual-mode Mn– and Sn–silica composites from millet husk: photoreduction of Cr(VI) and sulphanilamide, adsorptive capture of basic fuchsin." *Case Studies in Chemical and Environmental Engineering*, art. 101337, 2026.

CONFERENCE PAPERS & PRESENTATIONS

1. N. A. Biju, R. N. D., N. A. Athman, and **A. Ravi** (corresponding author). "AI-driven protein–protein interaction network analysis for diabetic wound healing." Bioradiance International Conference, 2026. ISBN 978-81-987828-6-1.
2. **A. Ravi** and T. George. "In-silico discovery of new Aurora Kinase-B inhibitors by 3D-QSAR and molecular docking studies." International Conference on Advanced Functional Materials (ICAFM 2018), S.B. College, Changanacherry, 2018.

FUNDED RESEARCH & GRANTS

1. **CSR Research Grant — Suven Pharmaceutical Ltd.** ₹81 lakhs; establishment and support of the Centre for Biomaterials and Sustainable Chemistry (CBSC), CVV. (2025)
2. **IKS Division, Ministry of Education, Government of India** — research project on natural adhesives. Co-Principal Investigator (Principal Investigator: Prof. Sunitha Grandhee). (2025)
3. **ATAL Faculty Development Programme (AICTE)** — ₹3.5 lakhs to conduct an ATAL FDP (Basic level). Coordinator. (2024)

Proposals under review

Co-Principal Investigator (with Dr. Arya R as PI) on a proposal for CBSC-CVV empanelment as a Technology Support Provider under the World Bank-assisted RAMP programme (Intervention K2), focused on AI-based image processing for quality control in Kerala's manufacturing MSMEs — currently under review.

TECHNICAL & LABORATORY SKILLS

Materials & synthesis

Hydrogel and 3D printed scaffold fabrication; silica-based molecular-sieve materials and functionalisation; nanocomposites; batch, autoclave and supercritical-fluid natural-product extraction.

Characterisation & analytical instrumentation

UV-visible and IR spectrophotometry; Raman spectroscopy; XPS; HR-TEM; SEM-EDX; elemental analysis (Computrace, ion-selective electrode); XRF; column, thin-layer and flash chromatography; in-vitro and antibacterial assays.

Computational & software

Molecular docking and 3D-QSAR (Schrödinger, AutoDock Vina); Gaussian, GaussSum, PyMOL, ChemDraw; machine-learning and graph-neural-network approaches for materials design and biological network analysis.

TRAINING & PROFESSIONAL DEVELOPMENT

- Six-month internship in Computer-Aided Drug Design (CADD) and 3D-QSAR — Drug Discovery Centre in collaboration with IIT Madras, Annamalai University (2015–16).
- Schrödinger software training (one month), IIT Madras.
- NPTEL certification — Analytical Chemistry (IIT Kharagpur, 8 weeks) and Composite Materials (IIT Kanpur, 12 weeks).
- Ten-day hands-on training in in-vitro biological analysis, VJ Biotech, Coimbatore.

ACADEMIC PROFILES

Google Scholar: scholar.google.com/citations?user=j30Or3cAAAAJ

ResearchGate: researchgate.net/profile/Athul-Ravi

IRINS / Vidwan: vidwan.inflibnet.ac.in/profile/552411

CVV Faculty Profile: cvv.ac.in/faculty/mr-athul-ravi

REFERENCES

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